

AYUSH KUMAR

FULL STACK DEVELOPER

7050184691 | ayushkumar44344@gmail.com | [LinkedIn](#) | [GitHub](#)

SUMMARY

Full Stack Developer and B.Tech undergraduate experienced in building full-stack web applications and backend systems. Skilled in React, Node.js, Express.js, MongoDB, AWS, Docker, Git, and GitHub.

TECHNICAL SKILLS

Languages: JavaScript, C++, Python, TypeScript

Frontend: React, React 19, Vite, HTML, CSS, Tailwind CSS

Backend: Node.js, Express.js, REST APIs

AI / ML: Google Gemini API, MediaPipe Pose, Ollama, Qwen3

Cloud / DevOps: AWS, Docker, Render, Git, GitHub

PROJECTS

-Synapse | P2P Skill-Swapping Platform

- Built a real-time P2P skill exchange platform using React 19, TypeScript, Node.js, Express, MongoDB, Redis, Socket.io, and Jitsi Meet with mutual skill matching and dynamic 1:1 meeting rooms.
- Engineered a virtual SKILL_CREDIT wallet, escrow ledger, and settlement workflow using MongoDB Transactions.
- Implemented room-based real-time, Zod validation, rate limiting, Multer/ImageKit uploads, and Swagger API documentation.

-MF (Mind & Fitness) | AI-Powered Fitness & Productivity Platform

- Developed a full-stack fitness and productivity platform with React, Node.js, Express.js, and MongoDB.
- Integrated Google Gemini API to power AI-driven nutrition analysis and Ustaadji, a personalized AI fitness coaching assistant.
- Developing an AI-powered exercise posture analysis system using Google's pretrained MediaPipe Pose model, with custom landmark-based geometry and exercise-specific rules for posture validation and rep tracking.
- Designed the posture-analysis architecture to combine computer vision-based pose estimation with rule-based form evaluation, with Ustaadji providing personalized feedback.

-AquaTrace | Chrome Extension

- Built a Manifest V3 Chrome extension that estimates the water, energy, and CO₂ footprint of AI conversations across ChatGPT, Claude, Gemini, and DeepSeek using local tokenization and research-based environmental estimates.
- Implemented local token counting with GPT Tokenizer (o200k_base), tracking prompts and responses separately per platform and model.
- Added daily water-usage budget alerts, footprint data export, persistent local statistics, and light/dark themes.

AquaTrace: <https://chromewebstore.google.com/detail/aquatrace/lmjgniffadnhnpjhjncpmaojpiflbcn>

-Personal AI Desktop Agent | In Progress

- Designed a self-learning application/folder resolver with tiered search and persistent JSON caching, cutting repeat lookups from full scans to instant retrieval.
- Integrated a locally hosted Qwen3 LLM via Ollama to convert natural-language requests into structured JSON actions with strict schema validation.
- Optimized LLM inference latency by ~10x through decoding-level JSON constraints, context/token tuning, and persistent model warm-loading.
- Implemented fuzzy, typo-tolerant, collision-safe name matching and a self-healing caching layer for portable path resolution.

OPEN SOURCE & ACHIEVEMENTS

- Summer of Code Fest (SOCF) — 2nd Place, Machine Learning Track
- CODEWARS — Finalist, 24-Hour National-Level Offline Hackathon organized by Hacker's Unity

EDUCATION

VIT Bhopal University — B.Tech, Cyber Security and Digital Forensics

2024 – 2028 | CGPA: 8.15